

ALARM ACKNOWLEDGEMENT — 18–19 NOVEMBER

HOW THIS LOG WORKS

In storm mode every alarm at console 1 must be acknowledged by the operator within **ninety seconds**. Unacknowledged, it escalates to the auto-dialler and wakes somebody at home. Acknowledgement is a physical keypress at the console. **The log is therefore a continuous record of whether a human being is in the chair.**

SUMMARY — 22:00 TO 06:00

Alarms raised	418
Acknowledged within 90 seconds	407
Median acknowledgement time	11 seconds
Escalated unacknowledged	11
Number of separate gaps in which this occurred	ONE

THE GAP

RAISED	ALARM	ACKNOWLEDGED	DELAY
02:38:44	Unit 1 — bearing temp high	03:05:12	26m 28s
02:41:03	Gate 2 — position feedback lag	03:05:16	24m 13s
02:43:57	Forebay — level rate of rise	03:05:19	21m 22s
02:47:12	Trash rack — differential rising	03:05:23	18m 11s
02:49:40	Unit 2 — governor oscillation	03:05:27	15m 47s
02:52:08	Drainage sump 3 — high level	03:05:31	13m 23s
02:54:33	Unit 1 — bearing temp high (repeat)	03:05:35	11m 02s
02:56:50	Crest lighting circuit — trip	03:05:39	8m 49s
02:58:19	Forebay — level rate of rise (repeat)	03:05:43	7m 24s
03:00:47	Gate 2 — position feedback lag (repeat)	03:05:46	4m 59s
03:03:11	Unit 2 — governor oscillation (repeat)	03:05:50	2m 39s

Eleven alarms, all acknowledged in a single burst of thirty-eight seconds beginning at 03:05:12. Eleven keypresses, one after another, at the rate of a person clearing a screen they have just come back to.

There is **no other unacknowledged escalation between 22:00 and 06:00**. Not one. The auto-dialler did not fire on the night; the escalations were cleared before it reached the call-out threshold at thirty minutes.

Operations note: the last acknowledgement before the gap was at **02:37:51**. Nothing was pressed at console 1 for the twenty-seven minutes that followed.

WHERE EVERY CARD WAS FOUND

CREDENTIAL	HOLDER	RECOVERED FROM	TIME
0001	A. SARN	On the deceased, clipped to belt	06:40
0004	H. VEY	On his person	08:05
0007	I. BALLARD	CONTROL ROOM — DESK, SECOND DRAWER	10:22
0022	D. OKONKWO	On her person	08:11
0031	C. MBEKI	On his person	08:09
V-118	P. WRENN	On his person	08:15

Credential 0007, issued to Ines Ballard, was recovered from the second drawer of the control room desk on a lanyard, at 10:22 on 19 November.

It was not on her body. It was not in her locker. It was not in her coat, her vehicle, or the site office.

Her body was recovered at 09:14 from the intake gallery, some 260 metres and two levels from the drawer her card was in.

NOTE BY SCENES OF CRIME

This item was photographed, bagged and scheduled on 19 November. It appears at line 31 of the property schedule attached to the original file. **It was not treated as significant at the time and no enquiry was made of it**, the investigation having by then settled on a sequence of events in which Ms Ballard was alive at 06:02.

Fingerprint examination of the card and lanyard: the surfaces are heavily overlaid and yielded no identifiable marks other than the holder's. Handling by a gloved person would leave nothing. The site issues gloves as standard winter PPE.

THE SPARE CONTROL ROOM KEY

DS The override key you were issued in September.

CM I gave it back on the eighteenth of September. Aurelio signed it in himself. It's two lines in the register and you've got the register.

DS Did you copy it?

CM Did I — no. Where would I copy a restricted-profile key? You need the card from the manufacturer.

Key register entry, return	18 SEP — SIGNED A. SARN
Physical key located	YES — SITE SAFE, TAGGED, UNUSED
Key profile	RESTRICTED — DUPLICATION REQUIRES OWNER CARD
Duplicate requests on file, three years	NONE
Control room door forced or picked	NO — EXAMINED, NO MARKS

The key was returned in September, is in the safe, and has nothing to do with this. The control room was entered and left by credential all night, and that record is already in front of you.

Turnaround: four working days.

WHEN THE DEBRIS ARRIVED

Differential pressure across the intake trash rack is monitored continuously. Debris loading obstructs flow and the differential rises. The trace is a direct record of what is sitting on the rack and when it got there.

TRACE — 19 NOVEMBER

TIME	DIFFERENTIAL (KPA)	INTERPRETATION
02:31	0.4	Gate 2 opens. Racks clear (raked 17 NOV).
02:50	0.5	Nominal.
03:12	0.5	Gate 2 closes. Flow ceases.
04:00	0.5	Static — no flow, no change.
04:12	3.8	STEP CHANGE. Peak inflow overtops into the gallery approach; a debris mat consolidates across panels 3-5.
05:00	4.1	Mat compacting.
07:00	4.1	Static.

COMPOSITION OF THE MAT

Bracken, spruce needles, birch twig, and one length of larch fence rail bearing a **Ravensgate Estate** stamp. The estate reported the loss of approximately eleven metres of boundary fencing from the north shore during the peak of the storm, between 04:00 and 05:00. The material in the mat is that fencing and what came down the hill with it.

The mat formed at 04:12. The deceased was underneath it.

She was against panel 4 before 04:12. Since the only flow into the gallery that night ceased at 03:12, and the racks were raked clear on 17 November, she reached the rack during the single period of flow and nothing has moved her since.

SECOND POST-MORTEM — REPORT

I have reviewed the original protocol, the retained samples, the photographic record and the body, and I concur with Dr Adeyemi in every particular.

QUESTION PUT	ANSWER
Cause of death	Drowning, freshwater. Confirmed.
Alive on entering the water	Yes. Confirmed by diatom distribution.
Injuries sustained before death	None. All trauma is post-mortem, from the rack.
Evidence of being pushed	None, and none would be expected. A push to the back leaves no mark on a body recovered from moving water two days later.
Time of death	I cannot give one, for the reasons the original report sets out at length, and I would decline to give one if pressed in court.

This examination adds nothing to the original. The limitation is not a failure of the first pathologist. It is a property of cold water, and no second opinion will move it. If the sequence of these two deaths is to be established, it must be established by something other than a body.

Turnaround: six weeks. Fee borne by the Review Unit.

ASHGROVE TECHNICAL SERVICES LTD

Registered	2018
Sole director and shareholder	M. VEY (spouse of H. Vey)
Registered office	THE VEY RESIDENTIAL ADDRESS
Employees	NONE
Clients, ever	ONE
That client	THENNICK INSTRUMENTATION LTD

RECEIPTS

INVOICE DATE	NARRATIVE ON THE INVOICE	AMOUNT	CAIRNHOLD CERTIFICATION SIGNED BY H. VEY
2019	"Technical advisory services"	\$61,000	6 days earlier
2021	"Technical advisory services"	\$70,000	9 days earlier
2022	"Technical advisory services"	\$55,000	4 days earlier
TOTAL		\$186,000	

Each payment follows, within nine days, a certification signed by Halloran Vey in favour of the paying company. No work product, correspondence, deliverable, timesheet or meeting record of any kind has been produced by either party in respect of the advisory services invoiced.

POSITION

Household mortgage, redeemed in full	2022, eleven days after the third receipt
Declared to Cairnhold under the conflict-of-interest policy	NEVER
Declared to RDSI	NEVER

CABINET C-14 — WHY THE DRAWING LIED

AS-BUILT SCHEDULE, C-14 — AMENDMENT HISTORY

REV	DATE	CHANGE	REVISER
A	1967	Original issue. C-14 shown DUAL-FED: F-5 (primary), F-3 (standby).	—
B	1994	Busbar rating update. Dual feed unchanged.	—
C	2019	F-3 standby feed REMOVED FROM DRAWING. C-14 shown SINGLE-FED from F-5. Reason given on the revision block: "legacy feeder decommissioned under RG-0419."	H. VEY

Revision C is the drawing held in the control room. It is the drawing an engineer consults before isolating. It is the drawing Aurelio Sarn was working from at 05:52 on 19 November.

RG-0419, SCOPE B — DECOMMISSIONING CERTIFICATE

Certificate title	PERMANENT DECOMMISSIONING — 11 kV FEEDER F-3
Declares	"ISOLATED, DE-TERMINATED AND PHYSICALLY REMOVED"
Certifying engineer	H. VEY
Goods-receipt countersignature	D. OKONKWO
Date	2019 — SAME DAY AS THE SCOPE A CERTIFICATE

PHYSICAL INSPECTION, 22 NOVEMBER

Feeder F-3 cable, turbine hall to substation	PRESENT AND INTACT
F-3 tie switch	PRESENT, OPERABLE, SCADA-CONTROLLABLE
F-3 termination at C-14 busbar	MADE OFF AND LIVE-CAPABLE
Evidence of any decommissioning work, ever	NONE

Feeder F-3 was certified removed, was never removed, and was erased from the drawing by the man who certified it. It stayed live, it stayed switchable from the control room, and it stopped appearing on the diagram that would have warned anybody it was there.

LEG TWO OF THE ORIGINAL FINDING

The original file records that "Ms Ballard and Mr Sarn were overheard at 02:25 in a serious disagreement." One person overheard it. This is that person, asked properly.

DS How did you hear an argument from the downstream powerhouse?

CM Somebody's handheld was keyed. It happens – you put a radio in a coat pocket and it presses against something and the whole net gets your conversation. It came over the powerhouse speaker for about forty seconds.

DS What did you hear?

CM Aurelio, angry. I know Aurelio's voice, he's got that flat way of getting quieter the crosser he gets. And another voice, also angry, and then it cut.

DS Whose was the second voice?

CM I told them Ines.

DS Did you recognise it as Ines?

 [Pause.]

CM No.

DS Then why did you say Ines?

CM Because she'd been on at Aurelio about the piezometers all evening and it was two in the morning and it was a gale and a voice came over the speaker and I thought, they're at it again. Nobody asked me if I recognised it. They asked me who was arguing and I said who I assumed.

DS Could it have been a man's voice?

CM Through a keyed handheld, in a storm, over a speaker in a machine hall? It could have been anybody. That's the honest answer and I should have given it the first time.

RADIO NET RECORDING

Recording retained	YES – 02:24:51 to 02:25:33
Voice A	IDENTIFIED – A. SARN
Voice B	NOT IDENTIFIABLE. Speech-comparison examination inconclusive as to identity and as to sex.
Intelligible content	"...never went in..." / "...first light..." / remainder unintelligible

Leg two of the original finding is an assumption made by a frightened twenty-six-year-old at two in the morning, recorded as an identification. Somebody argued with Aurelio Sarn at 02:25. The file cannot say who.

P. WRENN — THREE RECORDS

No single record accounts for this man's night. Three do.

(I) CONTROL ROOM DOOR

Entry	01:50:12
Exit	01:58:40
Any further entry before 07:00	NONE

He was not in the room the card was in, at any point after 01:58.

(II) RDSI INSTRUMENT VAN — DATALOGGER SESSION

TIME	OPERATOR ACTION AT THE VAN TERMINAL
02:20:14	Session opened, logging started, channel set
02:34:02	Manual mark entered – "gate 2 open, logging spill"
02:51:37	Channel gain adjusted
03:06:20	Manual mark entered – "level response nil, query"
03:29:55	Session closed

The van is on the dam crest, 300 m from the forebay parapet along an exposed walkway. Keystrokes at 02:34 and 02:51 bracket the window in which Ms Ballard entered the water.

(III) SATELLITE CALL RECORD

Number called	RDSI DUTY DESK
Connected	05:58:41
Cleared	06:14:09
Duration	15m 28s
Recording retained	YES — HIS VOICE CONTINUOUS THROUGHOUT
Handset location	RDSI VAN, DAM CREST

The F-3 tie was closed at 06:02:14 at console 1. He was on a recorded call from a van on the crest, six minutes' walk away, for the whole of that.

Take any one of these away and Piotr Wrenn is still a suspect. (i) leaves the forebay open. (ii) leaves 06:02 open. (iii) leaves 02:45 open. Together they close every window in this case, and only together.

He remains, on the Inspectorate's own account, guilty of accepting a data package he should have verified. That is a matter for a different tribunal.

CONTACT SURFACES — REPORT

EXAMINATION	RESULT
Arc pitting at the point of contact	Present, consistent with 11 kV
Tissue transfer	Present, matches the deceased
Evidence of tampering with the busbar	None
Evidence of a deliberately created fault path	None. The cabinet was in normal condition and did exactly what an energised cabinet does.
Tool marks, foreign metal, added conductor	None

Nothing was done to the cabinet. The cabinet did not need to be tampered with. Something was done to the *supply*, and metallurgy cannot see that. This examination confirms what B-1 and E-6 already told you.

Turnaround: five weeks.

THENNICK INSTRUMENTATION LTD

Status	TRADING
Contract RG-0419, invoiced to Cairnhold	\$1.41 M – PAID IN FULL, 2019
Purchase records for piezometers P-7 to P-12	NONE
Delivery notes to Ravensgate, 2019	CABLE AND JUNCTION BOX ONLY
Instrument manufacturer's serial allocation	NO INSTRUMENTS ALLOCATED TO THIS SITE, EVER
Drilling subcontractor engaged	NONE
Feeder decommissioning subcontractor engaged	NONE

Thennick installed a junction box and some cable, invoiced \$1.41 M for two scopes of work, and did neither of them. The instruments were never bought and no borehole was ever drilled. No electrical contractor was ever engaged to remove feeder F-3.

Directors interviewed under caution decline to comment. The Registry notes that Thennick has held eleven contracts with Cairnhold Power since 2018, all certified by the same Cairnhold engineer. Those contracts are now subject to a separate investigation which is outside the scope of this review.

FOREBAY PARAPET — REPORT

EXAMINATION	RESULT
Fibre recovery, parapet coping	Nil of evidential value
Contact traces, scuffing	Extensive and undatable. This is a working parapet.
Damage to handrail or coping	None
Footwear marks	None retained. The crest was under 30 mm of standing water and 90 km/h wind for eleven hours.
Biological traces	Nil

Storm Bettany washed this scene for eleven hours before anybody looked at it. There was never going to be anything here, and there is nothing. The examination cannot distinguish a fall from a jump from a push, and it cannot say that anybody went over this parapet at all.

Turnaround: three weeks. The Unit notes that this authority was worth considering and was not worth spending. Both of those things are true.